MUHAMMAD USMAN

Python Backend & AI Engineer
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SUMMARY

Python and AI engineer with 3 years of experience building data pipelines, RAG systems, LLM agents and MCP integrations for production. Currently leading a 3-person remote team at AAI Innovations (Germany), where I built the ingestion pipeline behind CORA, a regulatory compliance platform covering 250k+ regulations. Most of my work is on the backend of AI products: getting messy data in, making retrieval accurate, and keeping everything running reliably at scale.

EXPERIENCE

AAI Innovations GmbH | ***Software Engineer (Contract)*** **Feb 2025 - Present**

- Lead a remote pod of 3 engineers. I take requirements from stakeholders, split them into tasks for the team, review code and help people get unblocked so we deliver on time.
- Built the document ingestion pipeline for CORA (semantic chunking, image captioning, embeddings), taking processing from 1,500 to 225,000 regulations per month (150x).
- Deployed each stage as KEDA-autoscaled pods on Kubernetes, with MongoDB triggering the next phase, so the system scales up under load and back down when idle.

Maanz AI | ***Software Engineer*** **Aug 2024 - Feb 2025**

- Built and maintained Apache Airflow DAGs and Python services for high-volume data ingestion, transformation, and analytics on daily schedules, supporting reliable downstream data workflows
- Improved throughput and reliability across data workflows, cutting manual fixes and speeding up downstream jobs.
- Worked on Apache Airflow and Python data workflows for vehicle homologation and safety-critical automotive data, supporting correct and traceable results under strict quality requirements

Condoroo | ***Software Engineer (Part-time)*** **Aug 2023 - Aug 2024**

- Built the company's first customer-facing AI chatbot on WhatsApp (Botpress plus custom code). Customers verify who they are, then get live account details from Airtable or file complaints with no staff involved. Thousands still use it.
- Built Airtable automations and Node.js backend workflows for complex processes, including a document parsing pipeline, which streamlined document handling and supported more efficient operations

HIGHLIGHTED PROJECTS

CORA: AI Regulatory Compliance Assistant | ***AAI Innovations***

- AI agent platform for regulatory compliance, covering 250k+ regulations from the EU, US, UN, ISO and others.
- Built agentic workflows with LangGraph and Vertex AI, using LangSmith for tracing, debugging and evaluation.
- Designed RAG pipelines on MongoDB Atlas Search for accurate, low-latency retrieval across the full regulatory corpus.
- Created synthetic benchmark datasets, validated by domain experts, to measure retrieval and answer quality. The data I curated was also used to train the company's in-house models.

MCP Gateway | ***AAI Innovations***

- Built the MCP layer that lets customers plug specific product features into their own AI agents without buying the full product.
- One gateway hosts 5 MCP servers over the full Streamable HTTP protocol, working with clients like Claude and Cursor.
- Implemented OAuth 2.1 with Client ID Metadata Documents (CIMD), which replaces Dynamic Client Registration, plus per-request authorization.
- Tracks credit usage per user and routes each request to the existing product services, returning the results to the agent.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (Node.js), C, C++
AI & LLMs: LangGraph, Google ADK, MCP (Streamable HTTP), LangSmith, RAG, OpenAI and Anthropic APIs, Vertex AI, Hugging Face, PyTorch
Backend: FastAPI, Flask, Django, REST APIs, OAuth 2.1, pytest
Data & Storage: MongoDB (Atlas Search), PostgreSQL, pgvector, Pinecone, Chroma, Weaviate, Apache Airflow, ETL, web scraping
Infrastructure: Kubernetes, KEDA, Docker, AWS, GitHub Actions, CI/CD, Git

EDUCATION

B.E., Mechatronics Engineering **Nov 2020 - Jun 2024**
National University of Sciences & Technology (NUST), College of E&ME *Rawalpindi, Pakistan*